# INTRODUCTION TO LINEAR ALGEBRA. 29 May 2026

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**Linear Algebra**

- **Definition:** Linear algebra is the branch of mathematics that deals with vectors, vector spaces, linear transformations, and systems of linear equations.
- **Purpose:** It provides a framework to understand the properties and operations of these mathematical objects, which can be represented using matrices and vectors.
- **Importance:** It is a **foundational concept for Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Computer Vision (CV)**. Key concepts include scalars, vectors, matrices, mathematical operations on matrices, Eigenvalues, and Eigenvectors.

**Applications**

**1. Data Representation and Manipulation**

- **Datasets:** When creating a model, data is stored in vectors. Vectors become multidimensional as more features (parameters) are added to a dataset, with each feature representing one dimension of the vector.
- **House Price Predictor Example:** Features like Area, Rooms, and Location are **independent or input features**, while the Price is the **output or dependent feature**.

```
  Independent or Input Features                 Output or Dependent Feature
  Area          Rooms       Location                      Price
   |              |            |                            |
[ 1000            4           Mumbai ]  ---------------> [ 1.5cr ]
```

- These mathematical values—and even text (like "Mumbai")—are converted to numbers and stored in vectors.
- **Quantifying Relationships:** Mathematical operations quantify the relationship between parameters to give us the output. **Quantifying this relationship means measuring or expressing how strongly two or more variables are related using numbers** (which involves correlation, a part of statistics).
- **Model Logic:** These relations are built so the model can "think" about which parameter will affect the output and by how much. Just as human brains are trained to know that increasing a certain parameter will affect the output, the model is trained to do the same mathematically.
- Ultimately, linear algebra represents data in vectors and matrices and manipulates them through mathematical operations to analyse and extract insights.

**2. Machine Learning and Artificial Intelligence**

- **Model Training:** Relies heavily on linear algebra, specifically through matrix multiplication and matrix arithmetic operations.
- **Dimensionality Reduction:** Linear algebra works well with higher-dimensional data. Techniques like **Principal Component Analysis (PCA) use Eigenvalues and Eigenvectors to**

**reduce data from a higher dimension** (e.g., 500 dimensions/features) to a lower dimension (e.g., 2 or 3 dimensions) so it can be more easily understood.

- **Neural Networks:** Both forward propagation and backward propagation rely on matrix multiplication. Inputs are multiplied by the weights of hidden nodes in hidden layers using matrix arithmetic.

```
   Inputs                 Hidden nodes in hidden layer             Outputs
 (f1: Area)      ---- (w1, w2, w3) ---->  [ O ]
                                                         --------> [ O ]
 (f2: No. of     ---- (w4, w5, w6) ---->  [ O ]
      rooms)
```

**Matrix Multiplication Equivalent:**

- **Hardware and Libraries: GPUs perform faster because they have multiple cores that execute matrix multiplication in parallel**. Libraries like Tensor Flow convert matrix values into tensors to process them.

**3. Computer Graphics**

- Images, such as RGB images, are made up of pixels.

```
Image (RGB)        Pixels
  +---+---+---+
  |   |   |   |      => Scaling, Rotating, Black and white
  +---+---+---+
  |   |   |   |      => This all involves Linear Algebra
  +---+---+---+         ↳ Because of Transformations
```

- Modifying these images—such as scaling, rotating, and converting to black and white—all involves linear algebra because it relies on mathematical transformations.

**4. Optimization**

- Optimization involves solving mathematical models, such as linear equations.
- In algorithms like **Regression** (represented by the equation , where is the slope and is the intercept), the line of best fit on a graph (like predicting Prize vs. Size) is not drawn randomly.

```
 Prize
   |       .   /
   |     .   /   .
   |       /   .
   |     /   .
   |   / .
   | /_________________ Size

Equation: Y = mx + c   -> Regression Algorithm
              |    |
          Slope   Intercept
```

- It is mathematically solved to find the optimal values where the maximum fit is achieved and errors are minimal.